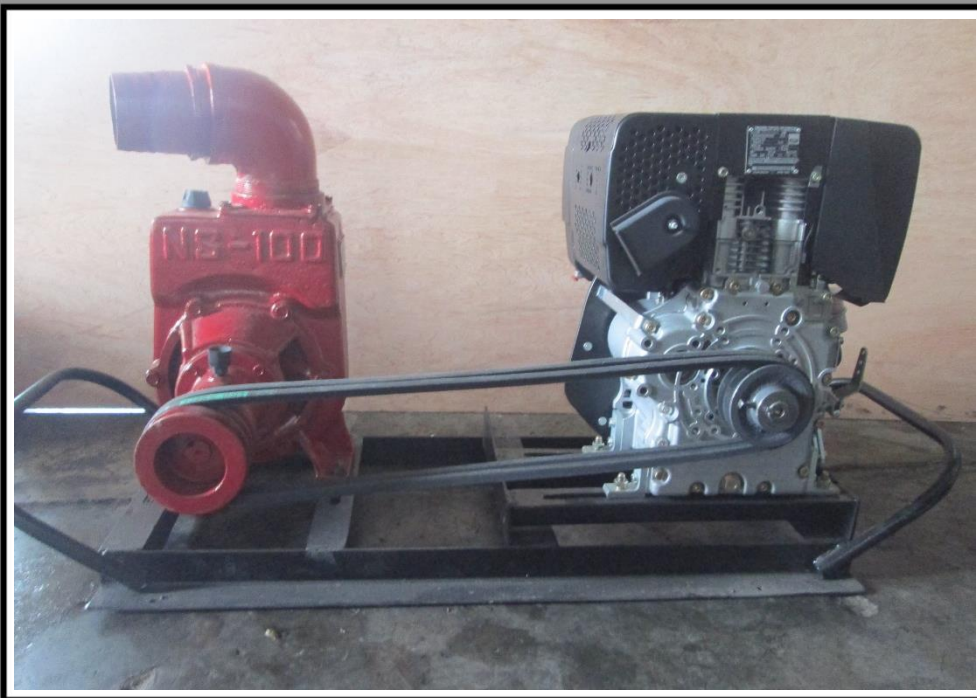




# AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

## TEST REPORT NO. 2014 - 154



**KITACO NS100 Centrifugal Pump coupled to  
HATZ 1B40 Diesel Engine**

## KITACO NS100 Centrifugal Pump coupled to HATZ 1B40 Diesel Engine

|                        |   |                                                                    |
|------------------------|---|--------------------------------------------------------------------|
| Test Report No.        | : | 2014 - 154                                                         |
| Date of Test           | : | October 26, 2014                                                   |
| Test Valid Until       | : | October 26, 2019                                                   |
| Brand                  | : | KITACO                                                             |
| Model                  | : | NS100                                                              |
| Serial No.             | : | No Data                                                            |
| Type and Size          | : | Single-end suction, 100 x 100 mm,<br>Self Priming Centrifugal pump |
| Manufacturer           | : | Made in China                                                      |
| Test Requested by      | : | M&W DISTRIBUTING CO., INC.<br>#14 Panay Ave., Quezon City          |
| Machine Classification | : | Commercial                                                         |

### SUMMARY

The KITACO NS100 Centrifugal Pump coupled to a HATZ 1B40 Diesel Engine was tested for performance and suction condition. It was tested at the recommended full throttle speed setting of the engine. The engine and pump both used double groove B-section pulleys with actual diameters of 102 mm (4 inches) and 114 mm (4.5 inches), respectively, and two pieces of V-belts. The engine and pump were mounted on a steel base.

The Maximum Total Head (H) was 27.0 m at zero discharge condition. The shaft speed of the pump and engine at this point were 2335 and 2764 rpm, respectively with Engine Fuel Consumption Rate of 2.17 L/h. The Maximum Discharge Capacity of 22.8 L/s occurred at 6.11 m Total Head with Fuel Consumption Rate of 1.86 L/h. The shaft speed of the pump and engine at this point were 1852 and 2208 rpm respectively. The Maximum System Efficiency of 10.7% occurred at 15.9 L/s Discharge Capacity.

The suction condition of the pump was also tested. With zero discharge, the Maximum Total Suction Lift was 8.37 m. The Net Positive Suction Head Available at this point was 1.63 m.

### PURPOSE AND SCOPE OF TEST

The purpose of the test was to establish the pump performance which included the Fuel Consumption Rate and the Total Head at varying discharge capacities and suction conditions. The pump was mounted on the laboratory test set-up shown in **Figure 1**.

The test was conducted on October 26, 2014 at the AMTEC Test Laboratory, UPLB, College, Laguna.



Figure 1. KITACO NS100 Centrifugal Pump coupled to HATZ 1B40 Diesel Engine on the test set-up

## METHOD OF TEST

The pump was tested following the “Philippine Agricultural Engineering Standards, PAES 115:2000, Agricultural Machinery – Centrifugal, Mixed-Flow, Axial-Flow Water Pumps – Methods of Test”.

## DESCRIPTION OF THE PUMP

The KITACO NS100 Self-priming Centrifugal Pump shown in **Figure 2**, was a single-end suction, 100 mm x 100 mm, centrifugal pump close-coupled to a 7.3 kW (9.78 Hp) HATZ 1B40 Diesel Engine. It was mounted on a steel base. The suction inlet and discharge outlet were threaded and designed for a 100 mm diameter pipe. The impeller was a semi-open type made of cast iron material connected to the pump shaft through a thread. It was housed in a cast iron volute casing enclosed in an aluminum chamber. The priming inlet was located on top of the pump beside the discharge outlet. The suction inlet was provided with a swing type one-way valve to retain the sufficient amount of liquid inside the chamber necessary to prime the pump. Located on top of the pump was the discharge outlet. The stuffing box used a mechanical seal.

The primemover, a 7.3 kW (9.78 Hp) HATZ 1B40 Diesel Engine was a four-stroke, air-cooled, single vertical cylinder diesel engine. Fuel feed system was of gravity type with the fuel tank located on top of the engine. It utilized a dry type air cleaner. Lubrication system was force-feed type. Cooling was done by air forced by a fan/flywheel encased in a metal shroud. It was started by a rope-recoil system. A muffler was used in the exhaust system. **Table 1** shows the detailed specifications of the pump.

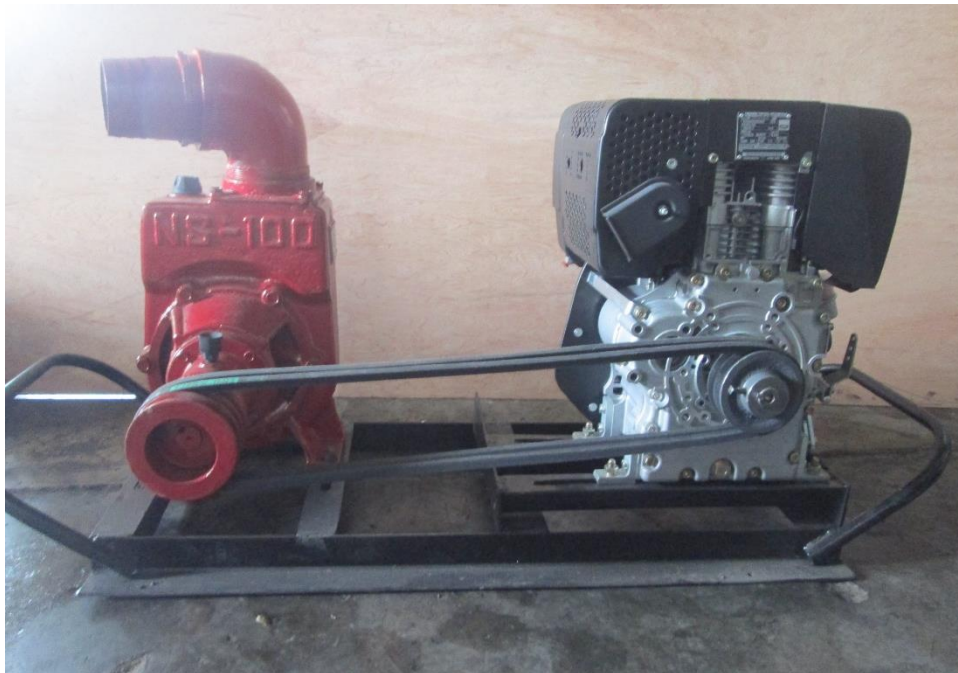


Figure 2. KITACO NS100 Self-priming Centrifugal Pump coupled to HATZ 1B40 Diesel Engine

### PUMPSET SPECIFICATIONS

Table 1. Detailed specifications of the KITACO NS100 Centrifugal Pump coupled to HATZ 1B40 Diesel Engine

|      |                                                                    |   |                                                   |
|------|--------------------------------------------------------------------|---|---------------------------------------------------|
| 1.   | Brand                                                              | : | KITACO                                            |
| 2.   | Model                                                              | : | NS100                                             |
| 3.   | Serial No.                                                         | : | No Data                                           |
| 4.   | Make                                                               | : | China                                             |
| 5.   | Type                                                               | : | Single-end suction, Self-priming centrifugal pump |
| 6.   | Size (mm)                                                          | : | 100 x 100                                         |
| 7.   | Overall Dimension and Weight (Including mounting frame and engine) |   |                                                   |
| 7.1  | Length (mm)                                                        | : | 1280                                              |
| 7.2  | Width (mm)                                                         | : | 450                                               |
| 7.3  | Height (mm)                                                        | : | 600                                               |
| 7.4  | Weight (kg)                                                        | : | 112                                               |
| 8.   | Impeller                                                           |   |                                                   |
| 8.1  | Type                                                               | : | Semi-open                                         |
| 8.2  | Diameter (mm)                                                      | : | 192                                               |
| 8.3  | Width (mm)                                                         | : | 36                                                |
| 8.4  | No. of vanes                                                       | : | 5                                                 |
| 8.5  | Material                                                           | : | Cast iron                                         |
| 9.   | Suction Inlet                                                      |   |                                                   |
| 9.1  | Type                                                               | : | Threaded flange                                   |
| 9.2  | Diameter (mm)                                                      | : | 100                                               |
| 9.3  | Material                                                           | : | Cast iron                                         |
| 10.  | Discharge Outlet                                                   |   |                                                   |
| 10.1 | Type                                                               | : | Threaded flange                                   |
| 10.2 | Diameter (mm)                                                      | : | 100                                               |
| 10.3 | Material                                                           | : | Cast iron                                         |

Table 1. (Cont'n)

|       |                                        |   |                                                                       |
|-------|----------------------------------------|---|-----------------------------------------------------------------------|
| 11.   | Casing                                 |   |                                                                       |
| 11.1  | Type                                   | : | Volute                                                                |
| 11.2  | Material                               | : | Cast Iron                                                             |
| 12.   | Stuffing box                           | : | Packing type                                                          |
| 13.   | Rotation                               | : | Counter clockwise when facing the suction inlet                       |
| 14.   | Rated speed (rpm)                      | : | 2000                                                                  |
| 15.   | Rated Maximum Discharge Capacity (L/s) | : | 28.3                                                                  |
| 16.   | Rated Maximum Total Head (m)           | : | 9.75                                                                  |
| 17.   | Primemover (based on Plate Rating)     |   |                                                                       |
| 17.1  | Brand                                  | : | HATZ                                                                  |
| 17.2  | Model                                  | : | 1B40                                                                  |
| 17.3  | Serial No.                             | : | 110181 049079                                                         |
| 17.4  | Make                                   | : | Germany                                                               |
| 17.5  | Type                                   | : | Four stroke cycle, air-cooled, single vertical cylinder diesel engine |
| 17.6  | Rated Output Power (kW)                |   |                                                                       |
|       | Maximum                                | : | 7.3 @ 3600 rpm                                                        |
|       | Continuous                             | : | No Data                                                               |
| 17.7  | Bore x Stroke (mm)                     | : | No Data                                                               |
| 17.8  | Displacement Volume (cc)               | : | 462                                                                   |
| 17.9  | Cooling System                         | : | Air-cooled                                                            |
| 17.10 | Lubricating System                     | : | Force-feed                                                            |
| 17.11 | Starting System                        | : | Rope-recoil                                                           |
| 17.12 | Air Cleaner                            | : | Dry type                                                              |
| 17.13 | Exhaust System                         | : | Muffler                                                               |



## PERFORMANCE TEST

Results of performance test are shown in **Table 2** and plotted in **Figure 3**.

Pumpset Tested : KITACO NS100 coupled to  
HATZ 1B40 Diesel Engine

Date of Test : October 26, 2014

Test Condition

Ambient Temperature (°C) : 29.2  
Atmospheric Pressure (mb) : 1013  
Pumping Water Temperature (°C) : 26.0

Table 2. Results of Performance Test

| ENGINE<br>SHAFT<br>SPEED<br>(rpm) | PUMP<br>SHAFT<br>SPEED<br>(rpm) | FUEL<br>CONSUMP-<br>TION<br>(L/h) | TOTAL<br>HEAD<br>(m) | DISCHARGE<br>CAPACITY<br>(L/s) | WATER<br>POWER<br>(kW) | INPUT<br>POWER<br>(kW) | SYSTEM<br>EFFICIENCY<br>(%) |
|-----------------------------------|---------------------------------|-----------------------------------|----------------------|--------------------------------|------------------------|------------------------|-----------------------------|
| 2764                              | 2335                            | 2.17                              | 27.0                 | 0                              | 0                      | 22.6                   | 0                           |
| 2731                              | 2320                            | 2.16                              | 24.8                 | 3.19                           | 0.78                   | 22.5                   | 3.46                        |
| 2644                              | 2238                            | 2.11                              | 23.0                 | 5.77                           | 1.30                   | 21.9                   | 5.94                        |
| 2559                              | 2164                            | 2.06                              | 20.7                 | 9.16                           | 1.86                   | 21.4                   | 8.68                        |
| 2510                              | 2116                            | 2.02                              | 18.5                 | 11.5                           | 2.08                   | 21.0                   | 9.89                        |
| 2467                              | 2071                            | 2.01                              | 16.4                 | 13.8                           | 2.22                   | 20.9                   | 10.6                        |
| 2417                              | 2026                            | 1.98                              | 14.2                 | 15.9                           | 2.21                   | 20.6                   | 10.7                        |
| 2349                              | 1967                            | 1.93                              | 11.5                 | 18.3                           | 2.08                   | 20.0                   | 10.4                        |
| 2272                              | 1905                            | 1.89                              | 9.02                 | 20.6                           | 1.82                   | 19.6                   | 9.26                        |
| 2208                              | 1852                            | 1.86                              | 6.11                 | 22.8                           | 1.37                   | 19.4                   | 7.05                        |



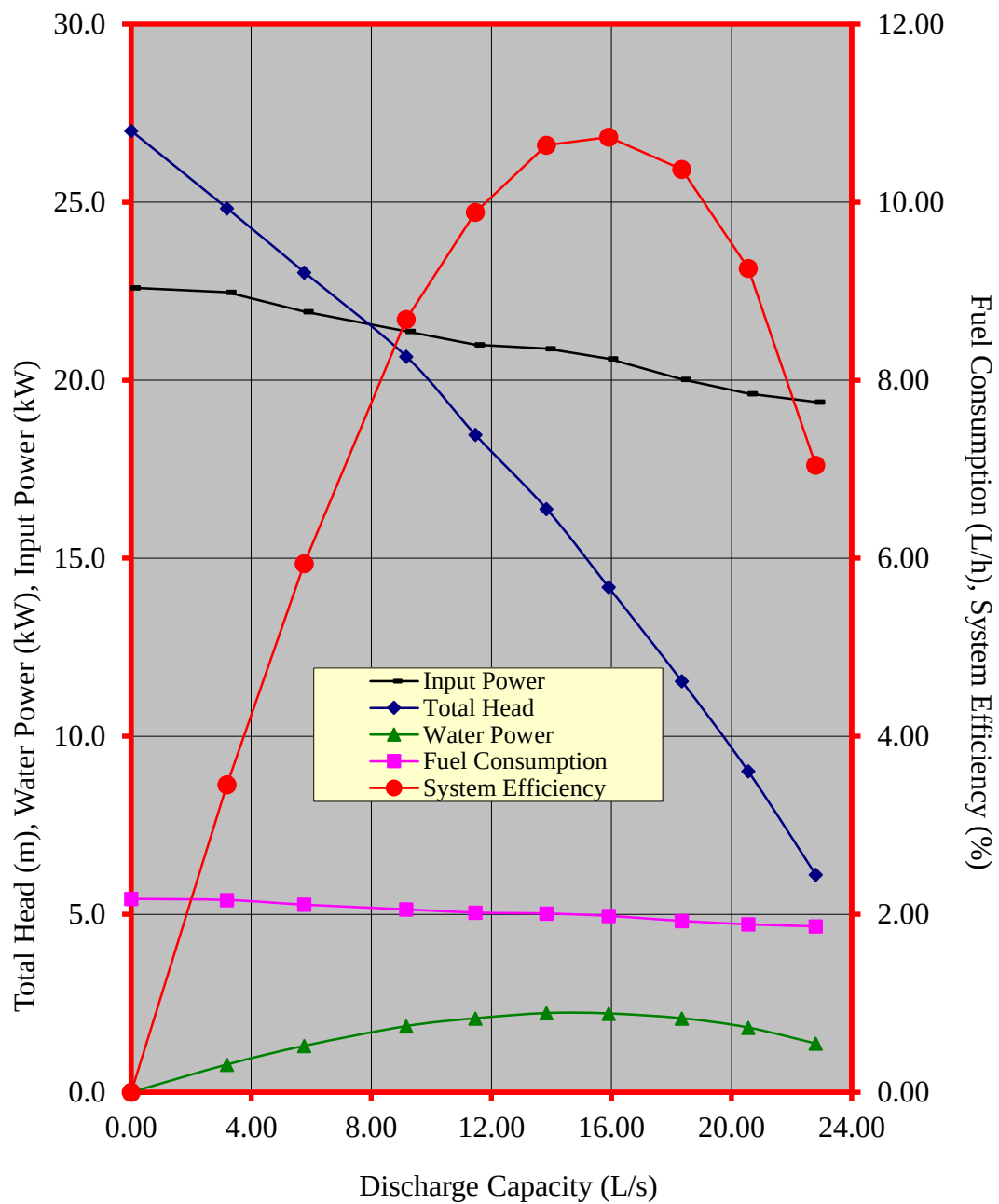


Figure 3. Performance characteristic curves of KITACO NS100 Centrifugal Pump coupled to HATZ 1B40 Diesel Engine

## CAVITATION TEST

Results of Cavitation Test are shown in **Table 3**.

|                                |   |                                                    |
|--------------------------------|---|----------------------------------------------------|
| Pumpset Tested                 | : | KITACO NS100 coupled to<br>HATZ 1B40 Diesel Engine |
| Date of Test                   | : | October 26, 2014                                   |
| Test Condition                 |   |                                                    |
| Ambient Temperature (°C)       | : | 29.5                                               |
| Atmospheric Pressure (mb)      | : | 1013                                               |
| Pumping Water Temperature (°C) | : | 26.0                                               |

Table 3. Results of Cavitation Test

| ENGINE<br>SHAFT<br>SPEED<br>(rpm) | PUMP<br>SHAFT<br>SPEED<br>(rpm) | TOTAL<br>SUCTION<br>LIFT<br>(m) | NET POSITIVE<br>SUCTION<br>HEAD<br>AVAILABLE<br>(m) | TOTAL<br>HEAD<br>(m) | DISCHARGE<br>CAPACITY<br>(L/s) |
|-----------------------------------|---------------------------------|---------------------------------|-----------------------------------------------------|----------------------|--------------------------------|
| 2213                              | 1859                            | 0.87                            | 9.13                                                | 6.31                 | 23.6                           |
| 2299                              | 1875                            | 2.07                            | 7.93                                                | 7.12                 | 22.5                           |
| 2328                              | 1900                            | 3.29                            | 6.71                                                | 8.24                 | 21.5                           |
| 2388                              | 1971                            | 4.75                            | 5.25                                                | 9.44                 | 19.3                           |
| 2551                              | 2137                            | 6.19                            | 3.81                                                | 10.3                 | 13.8                           |
| 2799                              | 2339                            | 8.37                            | 1.63                                                | 11.8                 | 0                              |

## DATA COMPUTATION

### ANNEX 1. Formula used during calculation and testing

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#### 1. Total Head (m), H

$$TDH = H_d - H_s$$

Where:  $H_d = \frac{P_2}{\gamma} + \frac{V_2^2}{2g} + Z_2$

$$H_s = \frac{P_1}{\gamma} + \frac{V_1^2}{2g} + Z_1$$

$H_d$  = Total Discharge Head (m)

$H_s$  = Total Suction Lift/Head (m)

$\frac{P_2}{\gamma}$  = Pressure gage reading on the discharge pipe at  
gage connection converted to meter

$\frac{P_1}{\gamma}$  = Vacuum gage reading on the suction pipe at the  
point of gage connection converted to meter

$\frac{V_2^2}{2g}$  = Velocity head on the discharge pipe converted  
to meter. Computed by determining the velocity  
of liquid on the pipe by formula,  $V = Q/A$  where  
 $V$ , velocity of liquid inside the pipe in meter per  
second,  $Q$ , Discharge rate of the pump in  
liters per second, and  $A$ , Cross-sectional area  
of the pipe in square meter.

$\frac{V_1^2}{2g}$  = Velocity head on the suction pipe converted to  
meter

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## Annex (Cont'n)

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$Z_2$  = The distance from the center of the pressure gage face to the centerline of the pump in meters

$Z_1$  = The distance from the center of the vacuum gage face to the centerline of the pump in meters

2. Discharge Capacity, Q, in liters per second (L/s) is determined through gravimetric method using a platform scale and a stopwatch.
3. Water Power, WP, is the output power of the pump in kilowatt (kW) and computed using:

$$WP = \frac{H \times Q}{102}$$

4. Input Power, IP, is the power on the input shaft of the pump. This is the power in kilowatt required to drive the pump and is determined by the formula:

$$IP = \frac{T \times N}{974}$$

Where:

T = Torque in kilogram-meter, on the input shaft of the pump and determined by a dynamometer

N = Input shaft angular speed in RPM

5. Pump Efficiency,  $\eta$ ,

$$\eta = \frac{WP}{IP}$$

6. Net Positive Suction Head Available, NPSHA. This is the net force (converted to meter) needed to push the water inside the pipe and computed by the formula:

$$NPSHA = P_{atm} - P_{vp} - H_s$$

Where:

$P_{atm}$  = Atmospheric pressure during the test converted to meters

$P_{vp}$  = Vapor pressure of the test liquid at certain temperature converted to meters

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N.B.

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